Collective Certified Robustness Against Graph Injection Attacks

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Introduction

Deep Graph Learning (DGL) has emerged as a crucial technique across various domains. However, recent studies have exposed vulnerabilities in DGL models, such as susceptibility to graph injection attacks (GIAs). Existing research only provides *sample-wise* certificates by verifying each node independently, leading to very limited certifying performance. In this paper, we present the first collective certificate against GIAs, which certifies a set of target nodes simultaneously.

Randomized Smoothing

For any node classifier $f(\cdot)$, its smoothed classifier $g(\cdot)$ is created by:

1. Node-aware Bi-Smoothing (Y. Lai. et. al. S&P2024):

$$\phi(G) = \left(\phi_e(G), \phi_n(G)\right);$$

- $\phi_e(G)$: randomly delete edges.
- $\phi_n(G)$: randomly delete nodes.

2. Majority vote: $g_v(G) \coloneqq \arg max_{y \in \{1,\dots,K\}}\ p_{v,y}(G)$,

$$p_{v,y}(G) \coloneqq P\left(f_v(\phi(G)) = y\right).$$

The goal of certified robustness is to verify :

$$g_v(G) \overset{?}{=} g_v(G'), \forall\, G' \in B_{\rho,\tau}(G).$$

Collective Certificate Problem

$B_{\rho,\tau}(G)$: the attacker can inject ρ malicious nodes, with τ malicious edges per node.

$\mathbb{T}$: A set of target nodes.

$$\min_{G' \in B_{\rho,\tau}(G)} \quad |\mathbb{T}| - \sum_{v \in \mathbb{T}} \mathbb{I}\{g_v(G') \neq g_v(G)\},$$

$$\text{s.t.} \quad |\tilde{\mathcal{V}}| \leq \rho,\ \delta(\tilde{v}) \leq \tau,\ \forall \tilde{v} \in \tilde{\mathcal{V}}.$$

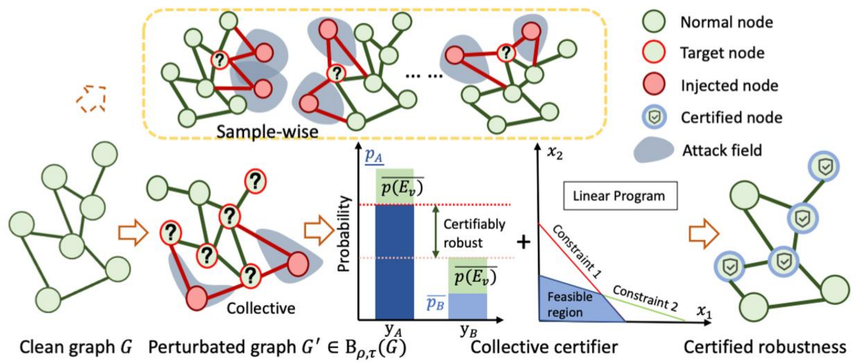

Figure 1: While the sample-wise certificate verifies target nodes one by one, our collective certificate verifies a set of target nodes simultaneously by linear programming.

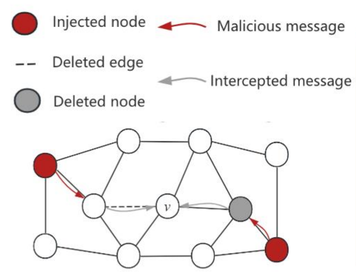

Figure 2: Malicious message has probability to be intercepted in the Node-aware Bi-smoothing (2-layer GNNs). (Adapted from Y. Scholten. et. al. NeurIPS 2022.)

Let $p(E_v)$ denotes the probability that the malicious messages is NOT intercepted, then we have that the absolute change of prediction probability is bounded by $p(E_v)$:

$$|p_{v,y}(G) - p_{v,y}(G')| \leq p(E_v).$$

Collective Certificate via Optimization

$p(E_v)$: the probability that the malicious messages is not intercepted.

$c_v \coloneqq p_{v,y*}(G) - max_{y \neq y*}\, p_{v,y}(G)$.

$$\max_{G' \in B_{\rho,\tau}(G)} \quad M = \sum_{v \in \mathbb{T}} \mathbb{I}\{\overline{p(E_v)} \geq c_v/2\},$$

$$\text{s.t.} \quad |\tilde{\mathcal{V}}| \leq \rho,\ \delta(\tilde{v}) \leq \tau,\ \forall \tilde{v} \in \tilde{\mathcal{V}},$$

The remaining $|\mathbb{T}| - M^*$ nodes are certified robust.

Sample-wise v.s. Collective Certificate

Sample-wise certificates (Bojchevski et al., 2020; Lai et al., 2023) essentially verify the prediction for a node one by one, assuming that the attacker can craft a different perturbed graph each time to attack every single node (Figure 1, top).

However, in reality, the attacker can only produce a *single* perturbed graph to **simultaneously** disrupt all predictions for a set of target nodes (Figure 1, bottom).

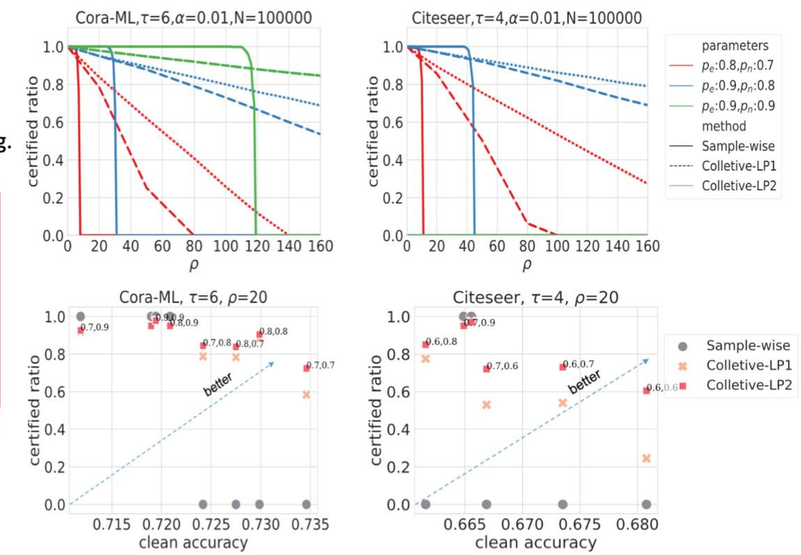

Figure 3: Certified ratio comparison (GCN model).

Evaluation Results

With the same smoothing parameter, both proposed collective certificates achieve a higher certifiable radius (Figure 3), outperforming the sample-wise approach significantly when the ρ is large. For example, in the Citeseer dataset, when ρ = 140, our Collective-LP1 and Collective-LP2 have the certified ratios of 73.0%, and 81.2%, while sample-wise can certify 0.0% nodes. Since the sample-wise and collective certificate share the same smoothed model, they always have the same clean accuracy.